

Why Measure Theory

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1 Introduction: Riemann Integration

1.1 Definitions

I will begin with the definitions of partitions, and upper and lower sums.

Definition 1 (Partition). Let $[a, b] \subset \mathbb{R}$ then P , a partition, is defined as

$$P = \{x_0, x_1, \dots, x_n\}$$

Such that

$$a = x_0 < x_1 < \dots < x_n = b$$

Note. P is a finite collection of points that subdivides the interval $[a, b]$. Importantly it is finite and partitions the domain of a function f on $\mathbb{R}$.

Definition 2 (Upper sums). Let f be a bounded function $f: [a, b] \rightarrow \mathbb{R}$, the upper sums of f for a partition P are defined as

$$U(f, P) = \sum_{k=1}^n (x_k - x_{k-1}) \sup_{[x_{k-1}, x_k]} f$$

Where $U(f)$ is defined as

$$U(f) = \inf_P U(f, P)$$

Definition 3 (Lower sums). Let f be a bounded function $f: [a, b] \rightarrow \mathbb{R}$, the lower sums of f for a partition P are defined as

$$L(f, P) = \sum_{k=1}^n (x_k - x_{k-1}) \inf_{[x_{k-1}, x_k]} f$$

Where $L(f)$ is defined as

$$L(f) = \sup_P L(f, P)$$

Now defining Riemann Integration.

Definition 4 (Riemann Integration). Let f be a bounded function $f: [a, b] \rightarrow \mathbb{R}$.

If for every $\epsilon > 0$ there exists a partition P_ϵ such that

$$U(f, P_\epsilon) - L(f, P_\epsilon) < \epsilon \tag{1}$$

then $f(x)$ is Riemann Integrable with

$$\int f dx = U(f) = L(f) \tag{2}$$

Notice from the definition that a function being Riemann Integrable relies on us finding a partition in which the upper and lower sums converge. In general how do we know before checking if a function will be Riemann Integrable?

1.2 Riemann Failure

1.2.1 Dirichlet Function

Let the sequence of numbers $\{r_1, r_2, \dots\}$ enumerate the rationals on $[0, 1]$.

Let $f_n(x)$ be a series of functions defined on $[0, 1]$ of the form

$$f_n(x) = \begin{cases} 1 : & x \in \{r_1, r_2, \dots, r_n\} \\ 0 : & x \notin \{r_1, r_2, \dots, r_n\} \end{cases} \quad (3)$$

1.2.2 Riemann Integration

I want to show each f_n is Riemann integrable. From the definition I must show

$$U(f_n, P_n) - L(f_n, P_n) < \epsilon$$

for every $\epsilon > 0$, for some partition P_n .

To begin fix $n \in \mathbb{N}$.

Take the partition P_n such that $(x_k - x_{k-1}) = \frac{\epsilon}{4n}$ for all consecutive pairs of points in P_n .

$$P = \{0, \frac{\epsilon}{4n}, \frac{2\epsilon}{4n}, \dots, 1\}$$

The total number of intervals in which there may exist rational numbers is $2n$; since there are n rationals where each can be in at most 2 intervals at once.

Note. if $r_k = x_k$ for instance, then $[x_{k-1}, x_k]$ and $[x_k, x_{k+1}]$ both contains r_k

From the definition of upper sums

$$U(f_n, P_n) = \sum_{k=1}^n (x_k - x_{k-1}) \sup_{[x_{k-1}, x_k]} f$$

The length of each interval as defined above is $(x_k - x_{k-1}) = (\frac{\epsilon}{4n})$, there exists at most $2n$ of these intervals, each with $\sup_{[x_k - x_{k-1}]} f = 1$.

All other possible intervals have a max combined length of 1, each with $\sup_{[x_k - x_{k-1}]} f = 0$.

This gives

$$U(f_n, P_n) = (\frac{\epsilon}{4n})(2n)(1) + (1)(0) = \frac{\epsilon}{2}$$

For the lower sums, since $\mathbb{I}$ is dense in $\mathbb{R}$, between any two real numbers there exists an irrational.

Meaning on any interval $[x_{k-1}, x_k]$, $\inf_{[x_{k-1}, x_k]} f = 0$.

This gives

$$L(f_n, P_n) = 0$$

This gives for any $\epsilon > 0$ for some $N \in \mathbb{N}$ whenever $n \geq N$

$$U(f_n, P_n) - L(f_n, P_n) = \frac{\epsilon}{2} < \epsilon$$

meaning the function is Riemann integrable with

$$\int_0^1 f_n dx = L(f_n) = 0$$

1.2.3 Limit

Letting $n \rightarrow \infty$ gives the limit of f_n as

$$\lim_{n \rightarrow \infty} f_n = \begin{cases} 1 : & x \in \mathbb{Q} \cap [0, 1] \\ 0 : & x \notin \mathbb{Q} \cap [0, 1] \end{cases} \quad (4)$$

Call this function f .

Now both $\mathbb{Q}$ dense in $\mathbb{R}$ as well as $\mathbb{I}$ dense in $\mathbb{R}$.

Meaning for any partition P , on any interval $[x_{k-1}, x_k] \subset [0, 1]$ there exists both a rational and irrational.

This gives $\inf_{[x_{k-1}, x_k]} f = 0$ and $\sup_{[x_{k-1}, x_k]} f = 1$ on every interval.

Which forces

$$U(f, P) = 1$$

and

$$L(f, P) = 0$$

for every partition P . Implying $U(f) = 1$ and $L(f) = 0$ which says that f is not Riemann integrable.

1.3 Lebesgue Explanation

1.3.1 Measure 0

To understand measure 0 first consider the concept of length.

Definition 5 (Length). The length of an open interval $(a, b) \subset \mathbb{R}$ is defined as

$$l((a, b)) = b - a \tag{5}$$

Now to define measure 0.

Definition 6 (Measure 0). A is a measure zero set if for every $\epsilon > 0$ there exists a countable covering $\bigcup_n O_n$ where each O_n is an open interval such that

$$A \subset \bigcup_n O_n \quad (6)$$

and

$$\sum_n l(O_n) \leq \epsilon \quad (7)$$

1.3.2 Lebesgue Criterion

By the Lebesgue criterion

Theorem 1 (Lebesgue Criterion). *A bounded function $f: [a, b] \rightarrow \mathbb{R}$ is Riemann integrable iff the set of discontinuities is measure 0*

We have now defined the class of Riemann integrable functions in terms of a concept of measure which did not previously exist in the context of Riemann integration.

1.3.3 Dirichlet Discontinuous Everywhere

Recall that

$$f = \lim_{n \rightarrow \infty} f_n = \begin{cases} 1 & x \in \mathbb{Q} \cap [0, 1] \\ 0 & x \notin \mathbb{Q} \cap [0, 1] \end{cases}$$

I want to show that f is discontinuous on all of $[0, 1]$.

First observe the definition of continuity.

Definition 7. A function f is continuous at a point c if for every $\epsilon > 0$ there exists a $\delta > 0$ such that whenever $|x - c| < \delta$ it follows that $|f(x) - f(c)| < \epsilon$

f Discontinuous

Begin by fixing $\epsilon \leq 1$, I will take $\epsilon_0 = \frac{1}{2}$. Choose any point $c \in [0, 1]$.

Again both $\mathbb{Q}$ and $\mathbb{I}$ dense in $\mathbb{R}$.

First consider $c \in \mathbb{Q} \cap [0, 1]$ then for any δ chosen there exists some irrational point

$$x \in (c - \delta, c + \delta) \implies |f(x) - f(c)| = 1 \geq \epsilon_0 \quad (8)$$

Consider the case $c \in \mathbb{I} \cap [0, 1]$ then for any δ chosen there exists some rational point

$$x \in (c - \delta, c + \delta) \implies |f(x) - f(c)| = 1 \geq \epsilon_0 \quad (9)$$

Meaning the set of discontinuous points of f is all of $(\mathbb{Q} \cap [0, 1]) \cup (\mathbb{I} \cap [0, 1]) = [0, 1]$ where $|[0, 1]| = 1$.

Note. I do not prove it here but it is proved in section 2.1

fn Measure 0 Discontinuities

To show that the set of discontinuities for every f_n is measure 0 first consider the set of points at which f_n is discontinuous, call it D_n .

For any points $c \in \{r_1, r_2, \dots, r_n\}$ since $\mathbb{I}$ dense in $\mathbb{R}$ we have for a fixed $\epsilon < 1$ and any $\delta > 0$

$$|x - c| < \delta \implies |f(x) - f(c)| = 1 \geq \epsilon$$

Making

$$D_n = \{r_1, r_2, \dots, r_n\}$$

For any points $c \notin \{r_1, r_2, \dots, r_n\}$ we can take $\delta = \frac{\min(|x-r_1|, |x-r_2|, \dots, |x-r_n|)}{2}$ since there are a fixed amount of rationals for any fixed $n \in \mathbb{N}$.

This would give us for

$$|x - c| < \delta \implies |f(x) - f(c)| = 0 < \epsilon$$

making f_n continuous on any points $x \notin \{r_1, r_2, \dots, r_n\}$.

Now consider the open covering $\bigcup_k O_k$ such that each

$$O_k = (r_k - \frac{\epsilon}{2^{k+1}}, r_k + \frac{\epsilon}{2^{k+1}})$$

Then

$$\begin{aligned} \sum_k l(O_k) &= \sum_k \frac{\epsilon}{2^k} \\ &= \frac{\frac{\epsilon}{2}}{1 - \frac{1}{2}} \\ &= \epsilon \end{aligned}$$

giving

$$\sum_k l(O_k) \leq \epsilon$$

making D_n measure 0 by definition.

We now understand why each f_n integrates but f itself fails to integrate. Our understanding lays not in the language of the Riemann integral itself but instead in the language of measure.

We also see that the method by which we partition the domain fails for functions that oscillate too quickly and will fail to be defined in spaces outside of $\mathbb{R}$.

1.4 Lebesgue Savior

Under mild assumptions, integrating with respect to measure can gives us

$$\lim_{n \rightarrow \infty} \int f_n d\mu = 0 = \int f d\mu \quad (10)$$

2 Outer Measure

Let $|\cdot|$ denote outer measure.

Definition 8 (Outer Measure). $|A| = \inf \sum_k l(O_k)$ Where inf is taken over all open coverings such that $A \subset \bigcup_k O_k$

2.1 Properties of Outer Measure

Theorem 2.

1. If $A \subseteq B$ then $|A| \leq |B|$
2. $|[a, b]| = b - a$
3. If $A_1, A_2, \dots$ satisfy $\bigcup_k A_k = B$, then $|B| \leq \sum_k |A_k|$
4. $|A + t| = |A|$

I will now prove each in order.

Proof: Monotonicity.

For monotonicity (1) first observe that since

$$A \subset B \subset \bigcup_k O_k$$

any open covering of B is an open covering of A .

Let

$$S_a = \left\{ \sum_k l(O_k) : O \text{ is an open cover of } A \right\}$$

Where each O is an open cover of the form $\bigcup_k O_k$.

Similarly let

$$S_b = \left\{ \sum_k l(O_k) : O \text{ is an open cover of } B \right\}$$

$S_b \subset S_a$ since any open covering of B is an open covering of A . It then follows $\inf S_a \leq \inf S_b$.

Giving $|A| \leq |B|$.

Note. Say $B \subset C$ then $\inf C \leq \inf B$; since any lower bound of C is a lower bound of B , making $\inf C$ a lower bound of B , where $\inf B$ is greatest lower bound.

□

Proof: $|[a, b]| = b - a$.

For the second property of outer measure begin by considering the open cover for $[a, b]$

$$(a - \epsilon, a + \epsilon) \cup (a, b) \cup (b - \epsilon, b + \epsilon)$$

Since $|[a, b]|$ is inf over all coverings we have

$$|[a, b]| \leq 2\epsilon + b - a + 2\epsilon$$

Where ϵ is arbitrary giving

$$|[a, b]| \leq b - a$$

For the other direction consider any open covering of $[a, b]$. I want to show that

$$\sum_k l(O_k) \geq b - a$$

hence giving

$$\inf \sum_k l(O_k) \geq b - a$$

Now $[a, b]$ is a compact interval, by Heine-Borel for every open cover

$$[a, b] \subset \bigcup_k O_k$$

there exists a finite subcollection such that

$$[a, b] \subset \bigcup_{k=1}^n O_k$$

Where

$$\sum_{k=1}^n l(O_k) \leq \sum_{k=1}^{\infty} l(O_k)$$

Since length is non negative.

Begin with $n = 1$ intervals. Any interval containing $[a, b]$ ie $[a, b] \subset O_k$ has length $l(O_k) \geq b - a$, since

$$(a, b) \subset [a, b] \subset O_k$$

Suppose the claim holds for n intervals.

Now consider $n + 1$ intervals covering $[a, b]$.

b must be contained in at least one of these intervals, call that interval (c, d) .

If $c < a$ then $l(c, d) \geq b - a$. So consider $a < c < b < d$. Since (c, d) covers none of $[a, c]$ we can invoke the inductive hypothesis on $[a, c]$ giving

$$\sum_{k=1}^n l(O_k) \geq c - a$$

for O_k covering $[a, c]$. Adding this to $l(c, d)$ gives

$$\begin{aligned} \sum_{k=1}^{n+1} l(O_k) &\geq (c - a) + (d - c) \\ &= d - a \\ &\geq b - a \end{aligned}$$

Completing the proof. □

Proof: Countable Subadditivity.

I want to show

$$|\bigcup_k A_k| \leq \sum_k |A_k|$$

Let

$$A = \bigcup_k A_k$$

Since outer measure is inf over all open covers, take for each A_k the open cover $O_k = \bigcup_i O_{ki}$ where

$$\sum_i l(O_{ki}) < |A_k| + \frac{\epsilon}{2^{k+1}}$$

Now since the O_k cover each A_k and $A = \bigcup_k A_k$

$$A \subset \bigcup_k O_k$$

Meaning

$$\begin{aligned} |A| &\leq \sum_k \sum_i l(O_{ki}) \\ &< \sum_k |A_k| + \frac{\epsilon}{2^{k+1}} \\ &= \sum_k |A_k| + \frac{\epsilon}{2} \end{aligned}$$

Which holds for all $\epsilon > 0$ giving

$$|\bigcup_k A_k| \leq \sum_k |A_k|$$

Completing the proof.

□

Proof: Translation Invariance.

I would like to show $|A + t| = |A|$.

First observe that

$$l((a, b)) = l((a + t, b + t))$$

since

$$l((a + t, b + t)) = (b + t) - (a + t) = b - a$$

Now observe that for any open cover of A call it O_A , O_{A+t} covers $A+t$.

Where O_{A+t} is defined as

$$\bigcup_k (O_k + t)$$

for $O_k \in O_A$.

Take any point in $A+t$ call it y . Then $y = x + t$ for some $x \in A$. Where $x \in O_k$ for some $O_k \in O_A$. Then $y \in O_k + t$. Where

$$l(O_k) = l(O_k + t)$$

Making the set of covers of A is a subset of the set of covers of $A+t$.

Let

$$S_A = \left\{ \sum_k l(O_k) : O \text{ is an open cover of } A \right\}$$

Where each O is an open cover of the form $\bigcup_k O_k$.

Similarly let

$$S_{A+t} = \left\{ \sum_k l(O_k) : O \text{ is an open cover of } A+t \right\}$$

Meaning since $|A|$ inf over all covers ie $\inf S_A$ and $S_A \subset S_{A+t}$, it follows $|A+t| \leq |A|$.

Now consider the sets $A+t+(-t)$ and $A+t$ where the left side is a translations of the right side. Giving

$$|A+t+(-t)| \leq |A+t| \implies |A| \leq |A+t|$$

Which follows from the above. Completing the proof $|A+t| = |A|$.

□

It seems that outer measure has all the desired properties we would want or need. The problem comes when we want to integrate with respect to measure. We would like to break up the set we are integrating over into disjoint sets whose union is the set itself. Therefore we would additionally want to see.

$$|\bigcup_k A_k| = \sum_k |A_k| \quad (11)$$

Which is not much different then our property of sub-additivity. This may seem intuitively true at first, but as we will see in a moment this is not actually true for all sets. Therefore we would like to be explicit about where and when the properties holds to be coherent in our application of measure theory.

2.2 Failure of Outer Measure as a Measure

2.2.1 Equivalence Classes

Definition 9 (Equivalence Relation). x is said to be equivalent to y denoted by $x \sim y$ iff $x - y \in \mathbb{Q}$

To be an equivalence relation it must satisfy the following properties

1. $x \sim x$
2. If $x \sim y$ then $y \sim x$
3. If $x \sim y$ and $y \sim z$ then $x \sim z$.

Proof: Equivalence Relation.

For $x \sim x$, notice for any value x , $x - x = 0$ and $0 \in \mathbb{Q}$.

For $x \sim y \implies y \sim x$, notice that $y - x = -1(x - y)$ where $\mathbb{Q}$ is closed under multiplication.

For $x \sim y$ and $y \sim z$, then $x \sim z$, notice $(x - y) + (y - z) = (x - z)$ where $\mathbb{Q}$ is closed under addition. $\square$

Definition 10 (Equivalence Class on $[0, 1]$). An equivalence class on $[0, 1]$ is a set of all equivalent elements $\{y : y \sim x\}$ for all $x \in [0, 1]$

Lemma 3. Any two equivalence classes on $[0, 1]$ are disjoint.

Proof: Equivalence Classes Disjoint.

Suppose for contradiction the intersection of two equivalence classes was non empty.

Let there be some element in A call it z and the same element in B . For arbitrary $x \in A$ and $y \in B$, $x - z \in \mathbb{Q}$ and $y - z \in \mathbb{Q}$. Let $x - z = u$ and $y - z = v$.

Solving for x and y gives $x = u + z$ and $y = v + z$. Now subtracting $x - y$, we get

$$x - y = (u + z) - (v + z) = u - v$$

But $u, v \in \mathbb{Q}$, where $\mathbb{Q}$ is closed under subtraction, giving $x - y \in \mathbb{Q}$. Which says $x \sim y$.

Taking any $x \in A$ gives $x \sim y$ for some $y \in B$ forcing $x \in B$. Making $A \subset B$. Similarly taking any $y \in B$ gives $y \sim x$ for some $x \in A$. Making $B \subset A$ or $A = B$.

Therefore any two equivalence classes on $[0, 1]$ are disjoint.

2.2.2 Vitalli

For the following argument I will use the Axiom of Choice to show that, countable additivity does not necessarily hold for all subsets of $\mathbb{R}$.

Axiom. For any collection of nonempty sets, there exists a function assigning to each set one of its elements.

To begin, let a set V contain a single element from each equivalence class on $[0, 1]$. The equivalence classes on the interval $[0, 1]$ are of the form $(\mathbb{Q} + x) \cap [0, 1]$. Making V uncountable.

Now consider the translations $V + q$ for $q \in [-1, 1] \cap \mathbb{Q}$.

Lemma 4. For any $p, q \in [-1, 1] \cap \mathbb{Q}$, the sets $V + p$ and $V + q$ are disjoint.

Proof: Translated Classes Disjoint.

To see this suppose some element w existed such that

$$w \in (V + p) \cap (V + q)$$

for $p, q \in \mathbb{Q} \cap [0, 1]$.

Then $w = v + p$ for some $v \in V$ and $w = u + q$ for some $u \in V$. So

$$v - w = p - q$$

making $v - u \in \mathbb{Q}$.

Meaning $v \sim u$, but V only contains one element from each equivalence class meaning $v = u$ forcing $p = q$. But that says the translated classes are the same, meaning each translated class must be disjoint. $\square$

Now I will show that the translated classes are bounded in measure but infinite summations of values; a contradiction.

Lemma 5 (Vitalli Bounded). $[0, 1] \subset \bigcup_q (V + q) \subset [-1, 2]$

Proof: Vitalli Bounded.

Take any arbitrary $x \in [0, 1]$.

x sits in some equivalence class since V contains some representative from that class.

Meaning $x - v = q$ for some $q \in \mathbb{Q}$ and some $v \in V$.

Since both $x, v \in [0, 1]$ it implies that $q \in [-1, 1] \cap \mathbb{Q}$. Since $\bigcup_q (V + q)$ is a union over all $q \in [-1, 1] \cap \mathbb{Q}$, we have

$$[0, 1] \subset \bigcup_q (V + q)$$

Now take an arbitrary $v \in V$ and $q \in [-1, 1] \cap \mathbb{Q}$. $v \in [0, 1]$ and $q \in [-1, 1]$ forcing $v + q \in [-1, 2]$.

Meaning

$$\bigcup_q (V + q) \subset [-1, 2]$$

Completing the proof. □

Now by monotonicity it must be the case that

$$|[0, 1]| \leq \left| \bigcup_q (V + q) \right| \leq |[-1, 2]| \tag{12}$$

Which implies

$$1 \leq |\bigcup_q (V + q)| \leq 3 \quad (13)$$

Since each $V + q$ consists of exactly the same amount of elements $|V + q| = |V + p|$.

Now suppose it is true for disjoint sets

$$|\bigcup_k A_k| = \sum_k |A_k| \quad (14)$$

then

$$|\bigcup_q (V + q)| = \sum_q |(V + q)| \quad (15)$$

Suppose that $|(V + q)| > 0$, since there are an infinite amount of $q \in [-1, 1]$ it would force $\sum_q |(V + q)| = \infty$.

Suppose then that $|(V + q)| = 0$, it would then force $\sum_q |(V + q)| = 0$.

Measure is non negative meaning $|(V + q)| \geq 0$, but either gives a contradiction to

$$1 \leq |\bigcup_q (V + q)| \leq 3 \quad (16)$$

Meaning for the vitalli set, (a valid subset of $\mathbb{R}$ from the Axiom of Choice)

$$|\bigcup_k A_k| = \sum_k |A_k| \quad (17)$$

does not hold.

Meaning it is not true in general and therefore we can not assert that

$$|\bigcup_k A_k| = \sum_k |A_k| \quad (18)$$

This as you will see later actually implies that outer measure is not a valid measure on all subsets of $\mathbb{R}$.

3 Integration on Measure

First I will define a what it means to be a measure and where exactly measures are defined.

Before that I will define a σ -algebra and measurable space.

Definition 11 (σ -algebra). A σ -algebra on a set X is a set of subset of X satisfying the following three properties.

1. $\emptyset \in S$
2. for any set $E \in S$, $X \setminus E \in S$
3. for any sequence of sets $E_1, E_2, \dots$ where each $E_k \in S$, $\bigcup_k E_k \in S$

We can quickly derive $X \in S$. First observe $X \setminus E$ is $\{x \in X : x \notin E\}$. Then from the first two properties $X \in S$.

It follows from De Morgan's laws that for any $E_1, E_2, \dots$ where each $E_k \in S$, $\bigcap_k E_k \in S$.

$$\begin{aligned}
E_k \in S &\implies E_k^c \in S \\
&\implies \bigcup_k E_k^c \in S \\
&\implies \left(\bigcup_k E_k^c\right)^c \in S \\
&\implies \bigcap_k E_k \in S
\end{aligned}$$

Where the last line follows from

$$\begin{aligned}
\left(\bigcup_k E_k^c\right)^c &= \left(\bigcap_k E_k\right)^c \\
&\implies \left(\bigcup_k E_k^c\right)^c = \left(\left(\bigcap_k E_k\right)^c\right)^c \\
&\implies \left(\bigcup_k E_k^c\right)^c = \bigcap_k E_k
\end{aligned}$$

Note. Notice that S is defined specifically such that it contains the sets for which we later want to apply countable additivity too.

I will now define a measure space.

Definition 12 (Measure Space). A measure space a set X and a σ -algebra defined on the set X denoted as (X, S) .

Now to define a measure and a measure space.

Definition 13 (Measure Space). Let (X, S) be a measurable space. μ is a measure on the measure space denoted by (X, S, μ) if $\mu(\emptyset) = 0$, and $\mu(\bigcup_k E_k) = \sum_k \mu(E_k)$ for disjoint E_k where each E_k in S .

Now we can define integration with respect to measure and prove our result from earlier.